Algorithmic Causal Dynamics: A Unified Framework for Emergence and Complexity in Relational Actualism

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Abstract

What does it mean for something to be more complex than something else? Standard dynamical systems theory has no physical answer: complexity is a description imposed on trajectories through a pre-existing phase space, not a property generated by the dynamics themselves. This paper provides a physical answer grounded in Relational Actualism (RA). We propose that the universe is a discrete, growing causal Directed Acyclic Graph (DAG), and that every tier of physical complexity—from fundamental particle interactions to biological cells to minds—is a structural feature of this graph, generated by the same underlying mechanism: the recursive coarse-graining of on-shell actualization events into stable causal closures.

We formalize this via a hierarchy of recursive coarse-graining operators (Π_n) mapping microscopic actualization events to macroscopic Structural Causal Models. This yields a precise topological definition of irreducible emergence: a subgraph emerges at scale n when it achieves sufficient causal closure to function as a single effective node at scale $n+1$. Applying this framework generates a four-tier continuum of physical complexity: (1) fundamental interactions, (2) dissipative structures, (3) autopoietic systems, and (4) agentic intelligence—each tier distinguished not by kind but by the depth and stability of its causal closure.

The result is a unified physical account of complexity in which chemistry, biology, and cognition are not separate domains requiring separate explanations, but successive levels of the same topological process. We derive the mathematics of each tier from first principles, grounding the Action Value function for agentic graph-steering in the Steering Cost Functional of RACI Theorem 5 and establishing the formal correspondence between the RA thermodynamic framework and Friston's Free Energy Principle. New result: the Tier 2/3 Causal Firewall threshold $\sigma = \lambda \tau_d \ell^3 = 1$ is now proved via the directed Erdős-Rényi theorem applied to the Poisson-CSG causal graph, resolving Hard Wall 1.

Keywords: Relational Actualism (RA) · Causal DAG · Irreducible Emergence · Coarse-Graining · Complexity · Autopoiesis · Active Inference · Graph Steering · Free Energy Principle

1 Introduction: Replacing the Phase Space with the Growing Graph

The question of what distinguishes a living cell from a crystal, or a mind from a thermostat, is one of the deepest in science. Standard accounts appeal to information processing, self-organization, or emergence—but these are descriptions of complexity, not physical explanations of it. They characterize what complex systems do; they do not identify the physical process that generates the hierarchy of complexity in the first place, or explain why that hierarchy has the specific tiers it does.

This paper offers a physical answer. Grounded in Relational Actualism (RA) (Sandeman, 2026a), we propose that every tier of physical complexity— from fundamental particle interactions to biological

cells to minds—is a structural feature of the universal causal Directed Acyclic Graph (DAG) $\mathscr{G} = (\mathscr{V}, \mathscr{E})$, generated by a single underlying mechanism: the recursive coarse-graining of on-shell actualization events into stable causal closures. Each vertex $v \in \mathscr{V}$ is an irreversible on-shell actualization event—a quantum transition in which a mediating boson crosses the kinematic threshold $p^\mu p_\mu = m^2$ (natural units throughout) and an irreversible causal fact is written into the structure of the universe. The macroscopic state space is not a pre-existing manifold; it is the accumulated causal history of these discrete events. The arena is continuously manufactured by the dynamics occurring within it.

This shift has a precise consequence for the theory of complexity. A physical system is not a point tracing a trajectory through an abstract phase space; it *is* a localized subgraph within the growing universal DAG. Its complexity is determined by the depth and stability of its causal closure: how thoroughly its internal dynamics are screened from its environment, and how many successive levels of coarse-graining are required to represent it as a single effective node at the next scale. Chemistry, biology, and cognition are not separate domains requiring separate explanations. They are successive tiers of the same topological process, each generated by the same formal mechanism applied at a higher level of causal closure.

The present paper develops this picture into a formal framework we call *Algorithmic Causal Dynamics* (ACD). The central contribution is the recursive coarse-graining hierarchy (Π_n) and its consequences:

Section 2 formalizes the coarse-graining projection Π_n mapping microscopic actualization events to macroscopic Structural Causal Models, grounding it in the Markov blanket isomorphism of RACI (Sandeman, 2026c). Section 3 establishes the recursive hierarchy and provides a rigorous topological definition of irreducible emergence—the first such definition grounded in causal graph physics rather than information-theoretic postulates. Section 4 derives the four-tier continuum of physical complexity: fundamental interactions, dissipative structures, autopoietic systems, and agentic intelligence, showing that each tier is distinguished by the depth and stability of its causal closure rather than by any qualitative difference in kind. Section 6 derives the mathematics of the highest tier in full—the Action Value function for agentic graph-steering—directly from the RA thermodynamic framework, unifying active inference with the Epistemic-Physical Equivalence bound of RACI Theorem 5 (Sandeman, 2026c) and establishing the formal correspondence with Friston's Free Energy Principle. Sections 7 through 9 state the empirical predictions, the relationship to the RA suite, and the open problems.

2 Formalizing the Coarse-Grained Causal Graph

2.1 The Microscopic Foundation

Let the fundamental universe be described by the microscopic causal DAG $\mathscr{G}_0 = (\mathscr{V}_0, \mathscr{E}_0)$, where every vertex $v \in \mathscr{V}_0$ is an irreversible on-shell actualization event as defined in RAQM (Sandeman, 2026a). The Local Ledger Condition (RAGC, Definition 1 (Sandeman, 2026b)) enforces energy-momentum conservation at every vertex:

$$\sum_{\substack{u \in \mathscr{V}_0 \\ (u,v) \in \mathscr{E}_0}} \mathbf{q}(u) = \sum_{\substack{w \in \mathscr{V}_0 \\ (v,w) \in \mathscr{E}_0}} \mathbf{q}(w), \tag{1}$$

where $\mathbf{q}(v) \in \mathbb{R}^n$ is the charge vector (energy-momentum, electric charge, baryon number, etc.) at v.

A macroscopic dynamical system does not compute every microscopic vertex. It operates on a coarse-grained projection of the fundamental graph into a macroscopic Structural Causal Model (SCM).

Definition 2.1 (Macroscopic SCM)**.** *A* macroscopic Structural Causal Model *at scale n is a DAG $\mathscr{G}_n = (\mathscr{V}_n, \mathscr{E}_n)$ together with:*

- Exogenous variables U*: the underlying, untracked microscopic actualization events (thermal bath, environmental noise).*
- Endogenous variables V*: the coarse-grained macroscopic states tracked at scale n.*
- *A structural equation $v_i = f_i(\mathrm{pa}(v_i), u_i)$ for each endogenous variable, where $\mathrm{pa}(v_i)$ are the causal parents of v_i in $\mathscr{G}_n$ and $u_i \in U$.*

2.2 The Coarse-Graining Projection

Definition 2.2 (Coarse-Graining Projection)**.** *The coarse-graining projection $\Pi_n : \mathscr{G}_{n-1} \to \mathscr{G}_n$ groups micro-histories of $\mathscr{G}_{n-1}$ into macroscopic equivalence classes. It is the repeated application, at successively larger scales, of the Step 5 hydrodynamic limit of the Engine of Becoming (Sandeman, 2026g): the same operation that produces the effective spacetime metric from the actualized graph in RAGC produces the effective biological and cognitive descriptions from the actualization graph in RAHC. The four tiers of the complexity hierarchy are four scales of the same coarse-graining.*

Two micro-histories $h_1, h_2 \in \mathscr{G}_{n-1}$ are equivalent under Π_n if and only if they produce indistinguishable probability distributions over future macroscopic observables at scale n:

$$h_1 \sim_n h_2 \iff P(\textit{observables} \mid h_1) = P(\textit{observables} \mid h_2). \tag{2}$$

The vertices $\mathscr{V}_n$ are the equivalence classes under $\sim_n$. The edges $\mathscr{E}_n$ represent the aggregate causal influence between these coarse-grained states, induced by the causal structure of $\mathscr{G}_{n-1}$.

The equivalence relation (2) is the formal content of the intuition that macroscopic descriptions discard microscopic detail that is irrelevant to future macro-observables. It is the RA analogue of the coarse-graining implicit in thermodynamics, but here applied to the causal graph directly rather than to a phase space distribution.

2.3 Connection to the Markov Blanket

The coarse-graining projection Π_n is not arbitrary: it is constrained by the causal structure of $\mathscr{G}_{n-1}$. Specifically, the boundary of the macroscopic equivalence class $[h]_n$ is determined by the Markov blanket of the subgraph being coarse-grained. By RACI Theorem 2 (Markov Blanket Isomorphism, (Sandeman, 2026c)), the Markov blanket $\mathrm{MB}(G_O)$ of an organism subgraph $G_O \subseteq \mathscr{G}$ is formally isomorphic to the boundary $\partial J^-(G_O)$ of its past light cone:

$$X_{G_O} \perp X_{\mathscr{V} \setminus (G_O \cup \partial J^-(G_O))} \,\Big|\, X_{\partial J^-(G_O)}. \tag{3}$$

This means that the coarse-graining boundary is not a matter of convention or epistemic choice; it is determined by the causal topology of the underlying graph. The macro-node in $\mathscr{G}_n$ that represents the coarse-grained system is precisely the node whose incoming and outgoing edges correspond to the system's Markov blanket.

3 The Recursive Coarse-Graining Hierarchy and the Topology of Emergence

3.1 Irreducible Emergence

In standard physics, "emergence" is often treated as an epistemic phenomenon: a property is emergent if it cannot be deduced from a tractable description of the parts. This epistemic account makes emergence a matter of computational convenience rather than physical reality.

Algorithmic Causal Dynamics rejects this. Because the universe is a growing graph and because the coarse-graining projection Π_n is constrained by the causal topology of the underlying structure, emergence has a topological definition.

Definition 3.1 (Irreducible Emergence)**.** *A subgraph $S \subseteq \mathscr{G}_{n-1}$ exhibits* irreducible emergence *at scale n if:*

(i) S achieves sufficient causal closure that its Markov blanket $\mathrm{MB}(S)$ *is stable under the growth dynamics of $\mathscr{G}_{n-1}$; and*

(ii) The macroscopic edges $\mathscr{E}_n$ incident on the coarse-grained node $\Pi_n(S) \in \mathscr{G}_n$ cannot be recovered from any strict sub-decomposition of S at scale $n-1$.

Condition (i) is the stability condition: the Markov blanket persists under continued actualization of the environment. Condition (ii) is the irreducibility condition: the causal influences that the emergent system exerts and receives at the macro scale cannot be recovered by examining its parts separately. This is not epistemic limitation; it is structural. The macroscopic edges $\mathscr{E}_n$ are real causal relationships that exist *only* at the coarse-grained level because they depend on the collective organization of the subgraph, not on any individual vertex.

3.2 The Recursive Hierarchy $(\Pi_1, \Pi_2, \ldots, \Pi_n)$

Applying the coarse-graining projection recursively generates a nested hierarchy of causal descriptions:

Level 0 ($\mathscr{G}_0$) The fundamental actualization events: on-shell boson exchanges satisfying $p^\mu p_\mu = m^2$.

Level 1 ($\Pi_1 \to \mathscr{G}_1$, *Chemistry*) A localized sub-network of Level 0 events forms a stable covalent structure. To the surrounding environment, the internal actualization history is screened off by the molecule's Markov blanket. The molecule acts as a single causal node in $\mathscr{G}_1$. This is the level at which the Assembly Index $A(M)$ (Murray et al., 2023) becomes the relevant complexity measure, bounded below by $A_{\mathrm{RA}}(M)$ per RACI Theorem 1 (Sandeman, 2026c).

Level 2 ($\Pi_2 \to \mathscr{G}_2$, *Cells*) A massive network of Level 1 molecular nodes achieves autopoietic closure: a Markov blanket that actively maintains itself against environmental perturbation. To the surrounding tissue, the internal metabolic actualization history is irrelevant; the cell acts as a single effective node in $\mathscr{G}_2$.

Level 3 ($\Pi_3 \to \mathscr{G}_3$, *Organisms*) A network of cells achieves organismal-level Markov closure. The organism acts as a node in $\mathscr{G}_3$ that interacts with other organisms and environmental structures.

Level 4 ($\Pi_4 \to \mathscr{G}_4$, *Agentic Intelligence*) An organism develops a *generative motif*—a persistent internal sub-DAG that is a homomorphic image of the environmental causal structure and runs predictions faster than the environment actualizes (Section 4.4.1). This is the level at which active inference (Friston, 2010) and graph steering (Section 6) operate.

Remark 3.2. The hierarchy is not a partition of natural kinds but a sequence of stable causal closure regimes. The relevant level for describing a system is determined by the scale at which its Markov blanket is stable. A system can participate in multiple levels simultaneously: a neuron is a Level 2 node (cell) that is also a component of a Level 3 node (organism) that steers at Level 4.

3.3 The End of Epistemic Reductionism

The topological definition of irreducible emergence has a sharp consequence for reductionism. Reducing a Level 4 agent to Level 0 atoms does not merely make the description intractable; it destroys the macroscopic edges $\mathscr{E}_4$ that govern the system's actual causal behavior. The agentic steering described in Section 6 operates exclusively over the macroscopic topology $\mathscr{G}_4$. No description at Level 0 or Level 1 recovers these edges, because they depend on the collective organization of the full subgraph at Level 4.

This is not a claim about computational complexity. It is a claim about causal structure: the edges $\mathscr{E}_n$ for $n \geq 2$ are physically real causal relationships that do not exist at lower scales. They are produced by the coarse-graining process, which is itself constrained by the causal topology of the underlying graph. Reductionism fails not because the details are too numerous to track, but because the relevant causal structure is not present at the microscopic level.

4 The Four-Tiered Continuum of Complexity

The recursive coarse-graining hierarchy (Π_n) is not a taxonomy imposed on nature from the outside; it is a physical structure generated by the causal topology of the universal DAG. Each tier in the following continuum corresponds to a specific regime of causal closure, characterised by the stability and depth of the Markov blanket achieved at that scale. The tiers are not discontinuous kinds but thresholds along a single continuous process—the progressive deepening of causal self-reference within the growing graph.

4.1 Tier 1: Fundamental Interactions (The Microscopic Engine)

At the base level $\mathscr{G}_0$, the universe expands via local, irreversible actualizations governed by the RA-CSG dynamics of RAGC (Sandeman, 2026b). When a mediating boson crosses the kinematic threshold $p^\mu p_\mu = m^2$, an irreversible vertex is inscribed. There is no coarse-graining here; this is the raw, uncompressed generation of causal history. The Local Ledger Condition enforces conservation at every vertex; the LLC is the discrete substrate from which all macroscopic conservation laws emerge in the continuum limit.

4.2 Tier 2: Dissipative Structures (Reactive Macroscopics)

Macroscopic dissipative structures—Rayleigh-Bénard convection cells, hurricanes, chemical oscillators—emerge when a subgraph of $\mathscr{G}_1$ provides an efficient topological pathway for entropy production across a thermodynamic gradient. They persist in time and possess a definable macroscopic SCM at Level 1.

The key distinction from autopoietic systems is the stability of the Markov blanket. A dissipative structure's "boundary" is a statistical pattern in $\mathscr{G}_1$, but it is not maintained by internal steering: the subgraph has no mechanism for detecting and correcting deviations from the boundary condition. When the thermodynamic gradient vanishes, the subgraph dissolves seamlessly into environmental noise. The criticality parameter σ—the ratio of internal actualization rate to environmental disruption rate (Sandeman, 2026c)— is non-zero but unregulated: it is set entirely by external conditions rather than by internal causal forcing.

Remark 4.1. Some dissipative structures (notably certain chemical oscillators) exhibit rudimentary self-repair under perturbation. The Tier 2 / Tier 3 boundary is therefore a continuum, not a sharp phase transition. The Causal Firewall conjecture of RACI (Sandeman, 2026c) identifies the critical threshold $\lambda \cdot \tau_d \cdot \ell^3 = 1$ as the candidate for a sharp boundary, pending the limit theorem discussed in that paper's hard wall.

4.3 Tier 3: Autopoietic Systems (The Edge of Chaos)

The emergence of biological life marks a phase transition in graph dynamics. An autopoietic system (Maturana and Varela, 1980) achieves a *stable Markov blanket*: a causal boundary $\partial J^-(G_O)$ that is actively maintained against environmental perturbation. By RACI Theorem 2 (Sandeman, 2026c), this boundary is formally isomorphic to the Markov blanket of the organism in the Bayesian network sense.

The defining property of Tier 3 is not merely the presence of a Markov blanket but the active regulation of the criticality parameter σ toward the edge of chaos ($\sigma \approx 1$). An autopoietic system continuously executes highly specific actualization sequences to maintain this regime:

- If $\sigma < 1$ (subcritical): internal actualization rate is insufficient to maintain the graph structure against environmental noise. The Markov blanket dissolves (death).
- If $\sigma > 1$ (supercritical): internal actualization rate exceeds the coherence threshold; the system loses the structured causal closure that distinguishes it from its environment (runaway growth, loss of functional identity).
- At $\sigma \approx 1$ (critical): the system maintains maximum computational and causal complexity compatible with stable Markov closure.

This maintenance is thermodynamically costly. By RACI Theorem 4 (Landauer Cost of Actualization (Sandeman, 2026c)), each actualization event that suppresses Δq bits of quantum uncertainty dissipates at minimum $W_{\text{act}} \geq \Delta q \cdot k_B T \ln 2$. Autopoietic persistence requires a continuous metabolic flux to pay this cost.

4.4 Tier 4: Agentic Systems (Active Inference and Graph Steering)

Intelligence represents the highest topological survival strategy available to a Tier 3 system. An agentic system extends its causal influence outward beyond its Markov blanket by maintaining an internal generative model of the external coarse-grained graph $\mathscr{G}_{\text{env}}$ and using that model to steer its own future actualization sequence.

4.4.1 The Topological Criterion for Agency: The Generative Motif

The formal distinction between Tier 3 and Tier 4 requires a precise topological criterion. A Tier 3 autopoietic system maintains its Markov blanket reactively: it detects perturbations and responds to them after they arrive. A Tier 4 agentic system predicts perturbations before they arrive by running physical simulations of the environment's future causal structure.

Definition 4.2 (Generative Motif)**.** *A subgraph $\mathscr{G}_{\text{model}} \subseteq G_O$ is a* generative motif *if:*

(i) $\mathscr{G}_{\text{model}}$ is a homomorphic image of $\mathscr{G}_{\text{env}}$ at the relevant coarse-grained scale: there exists a structure-preserving map $\phi : \mathscr{G}_{\text{env}} \to \mathscr{G}_{\text{model}}$ that preserves the causal partial order and the macroscopic edge structure $\mathscr{E}_n$;

(ii) The causal depth of $\mathscr{G}_{\text{model}}$ is sufficient to compute the probability distribution over future states of $\mathscr{G}_{\text{env}}$ before those states are actualized in the external graph; and

(iii) The simulation is executed via real, on-shell actualization events within G_O—the generative motif is a physical sub-process, not a disembodied computational abstraction.

Condition (iii) is crucial and requires unpacking.

4.4.2 The Thermodynamic Cost of Imagination

The generative motif is not floating in the off-shell potentia. Running a physical model of the environment requires real, on-shell actualization events within the agent's own graph — synaptic firings, ATP hydrolysis, ion channel dynamics in the biological case. These are irreversible causal events that write permanent vertices to the agent's internal DAG. This has a direct thermodynamic consequence:

Proposition 4.3 (Imagination Has a Landauer Floor)**.** *Each internal simulation step of the generative motif — each counterfactual evaluation of a future environmental state — is a forced actualization sequence π_{sim} that dissipates at minimum*

$$W_{\text{steer}}(\pi_{\text{sim}}) \;\geq\; k_B T \cdot \Delta F[q_{\text{sim}}], \tag{4}$$

where $\Delta F[q_{\text{sim}}]$ is the reduction in the agent's uncertainty about the future environmental state achieved by the simulation step.

Proof. Direct application of the Epistemic-Physical Equivalence bound (RAHC Theorem 5, Section 5.4) to the internal simulation sequence π_{sim}. □

Equation (4) has a direct empirical anchor: the human brain's consumption of approximately 20% of the body's resting metabolic energy (Attwell and Laughlin, 2001) is a physical consequence of maintaining a high-fidelity generative motif of the social and physical environment. This is not an evolutionary inefficiency; it is the Landauer floor for running a sufficiently detailed physical simulation of a complex world. The metabolic cost of intelligence scales with the fidelity and breadth of the internal model, not with the complexity of the actions it selects.

A further consequence: the agent cannot simulate arbitrarily many counterfactual futures simultaneously. The Action Value function (20) incorporates $W_{\text{steer}}(\pi)$ as the cost of both external actions *and* internal simulations that evaluate them. The finite metabolic budget of any physical agent therefore enforces a genuine trade-off between the depth of predictive simulation and the breadth of policy search — a thermodynamically grounded account of the bounded rationality observed in all biological intelligence (Kauffman, 2000).

Definition 4.4 (Tier 4 Transition Criterion)**.** *A system crosses the* Π_4 *threshold from Tier 3 to Tier 4 if and only if it contains a generative motif* $\mathscr{G}_{\text{model}}$ *whose internal recurrent causal structure (macroscopic edges* $\mathscr{E}_3$ *with sufficient loop density) can compute predictions of* $\mathscr{G}_{\text{env}}$ *faster than* $\mathscr{G}_{\text{env}}$ *actualizes—i.e., the internal simulation rate exceeds the environmental actualization rate at the relevant coarse-grained scale.*

The recurrence requirement—sufficient loop density in $\mathscr{E}_3$—is the topological signature of predictive capacity: recurrent causal loops enable the generative motif to propagate predictions forward in simulated time. A purely feedforward subgraph (no recurrent loops) can match the environment's current state but cannot anticipate its future states.

4.4.3 The Epistemic Nature of the Generative Motif's Content

A critical clarification is required to prevent a category error that theoretical neuroscience reviewers will correctly identify. The Generative Motif definition involves two distinct aspects that must be kept separate:

The substrate is ontic. The physical sub-DAG $\mathscr{G}_{\text{model}}$ is a real causal structure within the agent's own graph. Its actualization events — synaptic firings, ATP hydrolysis, ion channel dynamics — are irreversible on-shell events that pay the Landauer thermodynamic cost. The generative motif is not a mathematical abstraction; it is a physical sub-process. This is the sense in which condition (iii) of the Generative Motif definition is non-negotiable: the substrate is as ontic as any other part of the causal DAG.

The content is epistemic. The *content* of $\mathscr{G}_{\text{model}}$ — the causal relationships it represents about the environment — is an epistemic approximation, not an ontological replica. The homomorphism $\phi : \mathscr{G}_{\text{env}} \to \mathscr{G}_{\text{model}}$ in condition (i) operates at the coarse-grained macroscopic scale; it is a functional, lossy map, not a micro-physical isomorphism. The model is not the territory.

The "good enough" principle. Evolution does not select for truth; it selects for survival. The generative motif that persists is the one whose coarse-grained predictions are adequate to prevent Markov blanket breach — not the one that most faithfully represents the microscopic physics of the environment. A formally adequate generative motif need only satisfy:

(i) Causal-order preservation at the macroscopic scale: predictions respect the temporal structure of environmental actualization at the scale relevant to the agent's actions.
(ii) Sufficient fidelity for action selection: the probability distribution over future environmental states computed by $\mathscr{G}_{\text{model}}$ must be accurate enough to distinguish between actions that preserve versus threaten Markov closure.

A model that satisfies these two conditions is evolutionarily adequate regardless of whether it faithfully represents microscopic causal structure. This is why biological organisms maintain highly compressed, lossy world-models that are systematically wrong about physics at the quantum level yet remain extraordinarily effective at the ecological scale. The framework does not require biological organisms to be physicists; it requires them to be adequate predictors of ecologically relevant causal structure.

This distinction resolves a potential confusion about condition (i) of the Generative Motif definition. The homomorphism ϕ is not a claim that the organism's internal model achieves micro-physical accuracy. It is a claim about functional correspondence at the coarse-grained level: the macroscopic causal edges that govern survival-relevant environmental events are structurally represented in $\mathscr{G}_{\text{model}}$, even if the microscopic detail is massively compressed or absent. The physical substrate paying the Landauer cost is ontic; the representation it implements is epistemic.

This is the causal DAG formulation of active inference (Friston, 2010): the agent continuously minimizes the variational free energy between its internal generative motif and its sensory evidence by (i) updating the motif (perception) and (ii) selecting actions that steer the global DAG toward preferred states (action).

Section 6 derives the formal mathematics of graph steering from the RA framework.

4.4.4 Worked Example: The Three Levels in a Single Biological System

To prevent the category error identified above from arising in practice, we trace a single biological system—*E. coli* chemotaxis—through all three levels of the RAHC architecture simultaneously. This illustrates precisely how microscale on-shell events, coarse-grained causal nodes, and epistemic model content are related without being conflated.

Level 0 / Tier 1: The on-shell substrate. At the microscale, the physical reality of the chemotaxis system consists entirely of on-shell actualization events in the sense of RAQM (Sandeman, 2026a):

- **Ligand binding:** a maltose molecule undergoes an on-shell electromagnetic exchange with the MBP receptor binding pocket. This constitutes an irreversible actualization event that writes a vertex into the causal DAG and produces an environmental entanglement record.
- **Phosphorylation cascade:** CheA autophosphorylation and CheY-P production proceed via on-shell ATP hydrolysis events. Each hydrolysis event is a Kinematic Snap: a mediating boson (the γ-phosphate) crosses the mass shell, irreversibly transferring causal information down the signalling chain.
- **Flagellar motor switching:** CheY-P binding to the FliM switch complex triggers a conformational change via on-shell protein-protein interaction events, reversing flagellar rotation.

These are not metaphors. They are specific, catalogued on-shell events whose molecular mechanisms are known to $\sim$Ångström resolution. The substrate is ontic.

Level 1 / Tier 2–3: The coarse-grained causal node. The coarse-graining projection Π_2 groups the $\sim 10^5$ individual molecular actualization events per second into a single macro-node in $\mathscr{G}_2$: the *CheY-P gradient*. This node is not a summary statistic imposed by the observer; its boundary is fixed by the Markov blanket constraint (RAHC Theorem 3): the CheY-P gradient is the minimal coarse-grained state sufficient to predict the bacterium's future motor-switching behaviour, given the current environmental sugar gradient.

At Π_3, the bacterium's motility pattern (run-and-tumble) is a macro-node in $\mathscr{G}_3$. The causal edge from the sugar gradient to the motility pattern exists at the Tier 3 level and is *not present* in the Tier 0 description. It emerges from the collective organisation of the signalling subgraph—this is the irreducible emergence of Section 3.

Level 2 / Tier 3–4: The epistemic content. The bacterium does not have a Generative Motif in the full Tier 4 sense (it lacks the recursive predictive model of an agentic system). But it has a *proto-Generative Motif*: the CheY-P gradient constitutes a compressed, lossy causal representation of the external sugar gradient. This representation is:

- **Ontic substrate:** real phosphorylation events in a real sub-DAG, paying a real Landauer cost.
- **Epistemic content:** a functional, lossy map of the external environment, adequate to drive up-gradient motility without faithfully representing the microscopic distribution of individual sugar molecules.

The architectural summary. The reviewer who asks "how does the neural network implement graph-steering in terms of on-shell events?" is conflating Level 2 (epistemic content) with Level 0 (on-shell substrate). The answer is:

1. **The substrate** of every biological computation consists of on-shell events (ATP hydrolysis, ion channel openings, photon absorption, synaptic vesicle release). These are Tier 1 actualisations. Named explicitly in Section 4.4.3.
2. **The coarse-grained causal structure** at Tier 3–4 is real and causally efficacious but is not recoverable from the Tier 0 description alone. It requires the Π_n projection constrained by the Markov blanket.
3. **The epistemic content** of the internal model is a lossy functional map at the macroscopic tier. Demanding that it be specified in terms of Tier 0 on-shell events is a category error equivalent to demanding that Newton's gravitational law be expressed in terms of individual quark-gluon interactions.

The framework is not silent on the microscale implementation; it is precise about which questions belong to which level of description.

5 Foundational Theorems of Causal Complexity

The four-tier continuum of Section 4 is not merely a descriptive taxonomy. Each tier transition is governed by a precise mathematical condition, and each condition yields a theorem. This section establishes the five foundational theorems of causal complexity: the Assembly Index Correspondence (Tier 1/2 boundary), the DAG Markov Property and Markov Blanket Isomorphism (Tier 2/3 boundary), and the Landauer Cost and Epistemic-Physical Equivalence (Tier 3/4 boundary). These results are proved here; their biological applications are developed in the companion paper RACI (Sandeman, 2026c).

5.1 The Assembly Index Correspondence (Tier 1/2 Boundary)

The transition from Level 0 (fundamental interactions) to Level 1 (stable molecular structures) is governed by the irreversibility of covalent bond formation. The Assembly Theory framework of Murray et al. (Murray et al., 2023) quantifies molecular complexity by the minimum number of JOIN operations in a construction program. ACD provides the physical underpinning of this metric.

Definition 5.1 (Actualism Assembly Index)**.** *For a molecule M assembled in a region of the causal DAG $\mathscr{G}$, define the* molecular subgraph *G_M as the induced subgraph on the set of actualization events involved in the assembly of M. The Actualism Assembly Index is*

$$A_{\mathrm{RA}}(M) \;:=\; \mathrm{depth}(G_M) \;=\; \textit{length of the longest directed path in } G_M. \tag{5}$$

Theorem 5.2 (Assembly Index Correspondence)**.** *Under Condition C1′ (each JOIN operation corresponds to at least one irreversible actualization event, operationally defined as an entropy-producing on-shell energy transfer not reversed by subsequent unitary evolution) and Condition C2 (the ordering of operations is consistent with the causal partial order of G_M):*

$$A_{\mathrm{RA}}(M) \;\geq\; A(M), \tag{6}$$

with equality when G_M is a minimal assembly DAG.

Proof. The Assembly Index $A(M)$ counts the minimum number of JOIN operations. Each JOIN corresponds (by C1′) to at least one vertex in G_M. The minimum number of sequential such vertices, respecting the causal partial order (C2), is the depth of G_M. In a minimal DAG, depth equals the number of edges on the longest path $= A(M)$. In a non-minimal DAG (with redundant parallel paths), depth $\geq A(M)$ because parallel paths do not reduce the sequential minimum. □

Remark 5.3. Condition C1′ is established by a variational argument (Born-Oppenheimer approximation guarantees all stable covalent bonds are exothermic, mandating irreversible energy dissipation) and confirmed computationally by a B3LYP/6-311+G* density functional survey across the five primary bond types of biological assembly (C-C, C-N, C-O, S-S, P-O). The full treatment, including the polymer-regime coarse-graining rule and the linear scaling $A_{\mathrm{RA}}(M) \approx \kappa \cdot A(M)$ in condensed phases, is given in RACI (Sandeman, 2026c).

5.2 Coarse-Grained Assembly Depth for Metabolic Networks

The Assembly Index Correspondence (Theorem 5.2) characterises the causal depth of assembling a *single molecule* from elementary constituents. Living systems, however, are not single molecules: they are metabolic networks in which hundreds of molecules are produced by interconnected pathways. Applying A_{RA} directly to a cell would require enumerating all causal paths through the full intracellular DAG, which is computationally intractable for realistic systems. This section develops a network-level generalisation that is computationally tractable and yields a new class of predictions.

Definition 5.4 (Network critical path)**.** *Let $\mathscr{N} = (R, S, \prec)$ be a metabolic network, where R is the set of reactions, S the set of metabolite species, and $\prec$ the causal ordering induced by substrate-product relationships. A reaction $r \in R$ is* irreversible *if it satisfies Condition C1′: the net free energy change $\Delta G < 0$ under physiological conditions, ensuring the reaction constitutes at least one actualization event. The* critical path *$\mathscr{P}^* \subseteq R$ is the longest directed path through irreversible reactions from the network's primary input substrate to its essential output.*

Definition 5.5 (Coarse-grained RA assembly depth)**.** *For a metabolic network $\mathscr{N}$, the* coarse-grained RA assembly depth *is:*

$$\tilde{A}_{\mathrm{RA}}(\mathscr{N}) \;:=\; \mathrm{depth}(\mathscr{P}^*) \;=\; \textit{number of irreversible reactions on the critical path.} \tag{7}$$

Three properties follow directly from the definitions and Theorem 5.2:

Proposition 5.6 (Properties of $\tilde{A}_{\mathrm{RA}}$)**.**

(i) ***Lower bound:*** *$\tilde{A}_{\mathrm{RA}}(\mathscr{N}) \geq A(M^*)$, where M^* is the most assembly-complex essential product of $\mathscr{N}$. The network cannot shortcut below the Assembly Theory floor for its hardest product.*

(ii) ***Upper bound:*** *$\tilde{A}_{\mathrm{RA}}(\mathscr{N}) \leq A_{\mathrm{RA}}(M^*)$. The network finds shorter causal paths to M^* than isolated single-molecule synthesis by reusing intermediates.*

(iii) ***Selection pressure:*** *Under replication competition, organisms with smaller $\tilde{A}_{\mathrm{RA}}(\mathscr{N})$ replicate faster, since a shorter critical path corresponds to fewer sequential bottlenecks in the cell cycle. Natural selection acts to minimise $\tilde{A}_{\mathrm{RA}}$.*

Proof. Property (i) follows because the critical path must pass through the production of M^*, and Theorem 5.2 requires at least $A(M^*)$ irreversible steps to assemble M^*. Property (ii) follows because the network can reuse earlier intermediates as sub-assemblies, reducing the number of fresh sequential steps relative to de novo synthesis (with equality when no intermediate on the critical path to M^* is shared with a shallower network route). Property (iii) follows because the cell cycle time is bounded below by the critical path length of the metabolic network: a cell cannot complete division faster than it can complete its slowest irreversible sequential dependency. □

5.2.1 Worked Example: Glycolysis

Glycolysis — the ten-step conversion of glucose to pyruvate — is the universal core metabolic pathway and provides a worked example of $\tilde{A}_{\mathrm{RA}}$ that can be compared against the Assembly Theory toolkit. The pathway has the following structure relevant to the critical path analysis (Meyerhof, 1945):

Step graph. Let $G_{\mathrm{glyc}} = (V_{\mathrm{glyc}}, E_{\mathrm{glyc}})$ where vertices are the ten enzymatic reactions and directed edges point from substrate to product. Three reactions are *strongly irreversible* under physiological conditions (i.e., satisfy C1′ with $\Delta G \ll 0$):

- r_1: Glucose $\to$ Glucose-6-phosphate (hexokinase, $\Delta G \approx -16.7$ kJ/mol)
- r_3: Fructose-6-phosphate $\to$ Fructose-1,6-bisphosphate (PFK-1, $\Delta G \approx -14.2$ kJ/mol)
- r_{10}: Phosphoenolpyruvate $\to$ Pyruvate (pyruvate kinase, $\Delta G \approx -31.4$ kJ/mol)

The remaining seven reactions are near-equilibrium ($|\Delta G| < 5$ kJ/mol) and are thermodynamically reversible under physiological conditions; they do not satisfy C1′ and are not counted as committed actualization steps.

Branching structure. At reaction r_4 (aldolase), fructose-1,6-bisphosphate is cleaved into two 3-carbon species (DHAP and G3P), which are interconverted by triose phosphate isomerase and then proceed in parallel through reactions r_6–r_{10}. The causal DAG therefore has a *fork* at r_4 and a *join* at the output (two molecules of pyruvate). The critical path runs through the longest sequential chain of irreversible steps:

$$\mathscr{P}^*_{\text{glyc}} = r_1 \to r_3 \to r_{10}, \quad \tilde{A}_{\text{RA}}(\text{glycolysis}) = 3. \tag{8}$$

Interpretation. The coarse-grained assembly depth of glycolysis is 3: there are three sequential irreversible commitments between glucose input and pyruvate output, with the intermediate seven reactions forming a near-equilibrium buffer that can run in either direction. This has a concrete biological meaning: the rate-limiting bottleneck of glycolysis is not any single enzyme's catalytic speed but the three points of irreversible commitment — the two ATP-investment phosphorylations and the final energy-recovery step. Allosteric regulation of glycolysis concentrates at exactly these three enzymes (hexokinase, PFK-1, and pyruvate kinase), consistent with the prediction that selection pressure acts at the irreversible nodes.

Comparison with full A_{RA}. For isolated pyruvate synthesis from scratch, $A_{\text{RA}}(\text{pyruvate})$ would count every actualization event involved in assembling the pyruvate molecule from atoms — a much larger number, bounded below by $A(\text{pyruvate})$ as established by Theorem 5.2. The glycolytic network achieves $\tilde{A}_{\text{RA}} = 3$ by leveraging the pre-assembled glucose molecule as a starting point and routing most of the transformation through reversible near-equilibrium steps, confirming Property (ii) of Proposition 5.6.

5.2.2 Prediction: Cross-Organism Scaling

Proposition 5.6(iii) generates a testable prediction:

Conjecture 5.7 (Evolutionary minimisation of $\tilde{A}_{\text{RA}}$). Across evolutionary lineages, organisms should exhibit systematic reduction in $\tilde{A}_{\text{RA}}(\mathscr{N})$ for their core metabolic pathways over evolutionary time, independently of thermodynamic pathway efficiency. The ratio

$$\rho_{12} = \frac{\tilde{A}_{\text{RA}}(\mathscr{N}_1)}{\tilde{A}_{\text{RA}}(\mathscr{N}_2)} \tag{9}$$

for two organisms $\mathscr{N}_1$, $\mathscr{N}_2$ should correlate with their phylogenetic distance and metabolic complexity, measurable with standard KEGG (Kanehisa & Goto, 2000) or BiGG metabolic network databases.

This conjecture is testable immediately using existing data: the KEGG metabolic database contains curated pathway graphs for thousands of organisms, each reaction annotated with thermodynamic reversibility data. Computing $\tilde{A}_{\text{RA}}$ for core pathways (glycolysis, the citric acid cycle, fatty acid synthesis) across organisms spanning the phylogenetic tree is a graph-algorithmic computation requiring no new experimental data. The prediction is that organisms with more ancient metabolic networks (archaea, early-branching bacteria) will exhibit higher $\tilde{A}_{\text{RA}}$ for equivalent pathways than their more derived counterparts, reflecting evolutionary pressure toward shorter irreversible commitment chains.

5.3 The Markov Blanket Isomorphism (Tier 2/3 Boundary)

The transition from Level 1 (chemistry) to Level 2 (autopoietic cells) requires a precise topological criterion for causal closure. The Markov blanket isomorphism establishes that this criterion is the

equivalence between the organism's statistical boundary (the Markov blanket) and its physical boundary (the past light cone boundary in the causal DAG).

In a Bayesian network $N = (X, E_N)$, the Markov blanket $\mathrm{MB}(S)$ of a node set S is the minimal set rendering S conditionally independent of all other nodes:

$$X_S \perp X_{V \setminus (S \cup \mathrm{MB}(S))} \mid X_{\mathrm{MB}(S)}. \tag{10}$$

In the RA causal DAG, define the past light cone $J^-(v) := \{u \in V : u \text{ causally precedes } v\} \cup \{v\}$ and its boundary $\partial J^-(v) := \{u \in J^-(v) : \exists w \notin J^-(v) \text{ with } (u,w) \in E\}$.

Theorem 5.8 (DAG Markov Property)**.** *In the causal DAG $\mathscr{G}$ with the RA-CSG probability measure, for any vertex v:*

$$\rho_v \perp \rho_{V \setminus J^-(v)} \,\Big|\, \rho_{\partial J^-(v)}. \tag{11}$$

Proof. The state at v is determined entirely by its causal predecessors $J^-(v)$. The boundary $\partial J^-(v)$ screens off all further causal information: any state $u \notin J^-(v)$ can influence v only through paths passing through $\partial J^-(v)$. The conditional independence follows from the d-separation criterion on DAGs (Pearl, 1988): v is d-separated from $V \setminus J^-(v)$ by $\partial J^-(v)$. □

Theorem 5.9 (Markov Blanket ↔ Light Cone Boundary Isomorphism)**.** [Lean-verified core.] *Let O be an organism modelled as a subgraph $G_O \subseteq \mathscr{G}$. There exists a coarse-graining map $\varphi : \partial J^-(G_O) \to \mathrm{MB}(G_O)$ such that:*

(a) *Sensory states correspond to incoming boundary vertices of $\partial J^-(G_O)$.*
(b) *Active states correspond to outgoing boundary vertices of G_O.*
(c) *The conditional independence structure is preserved:*

$$X_{G_O} \perp X_{V \setminus (G_O \cup \partial J^-(G_O))} \,\Big|\, X_{\partial J^-(G_O)}. \tag{12}$$

Proof. Parts (a) and (b) are definitional correspondences. For (c): by Theorem 5.8, ρ_v is conditionally independent of $V \setminus J^-(v)$ given $\partial J^-(v)$ for each $v \in G_O$. Since $\partial J^-(G_O) \subseteq \partial J^-(v)$ for all $v \in G_O$, joint conditional independence follows by the intersection property of conditional independence on DAGs. □

5.4 The Thermodynamics of Causal Forcing (Tier 3/4 Boundary)

The transition from Level 3 (autopoietic organisms) to Level 4 (agentic intelligence) requires that the system not merely maintain causal closure but actively force the growth of the DAG toward preferred states. This forcing carries a thermodynamic cost that bounds all intelligent behaviour.

Theorem 5.10 (Landauer Cost of Actualization)**.** [Lean-verified.] *Each irreversible actualization event that eliminates Δq bits of quantum uncertainty dissipates at minimum*

$$W_{\mathrm{act}} = \Delta q \cdot k_B T \cdot \ln 2. \tag{13}$$

Proof. By Landauer's principle (Landauer, 1961): the von Neumann entropy decrease $\Delta S_{\mathrm{vN}} = \Delta q \cdot \ln 2$ of the system must be transferred to the environment as heat $\geq T \Delta S_{\mathrm{vN}} = k_B T \cdot \Delta q \cdot \ln 2$. □

Definition 5.11 (Steering Cost Functional)**.** *For a forced actualization sequence $\pi = (v_1, \ldots, v_n)$:*

$$W_{\mathrm{steer}}(\pi) := \sum_{k=1}^{n} \Big[W_{\mathrm{act}}(v_k) + W_{\mathrm{sel}}(v_k) \Big], \tag{14}$$

$$W_{\mathrm{sel}}(v_k) = k_B T \cdot \ln \left(\frac{|\mathscr{A}\mathscr{P}_{\mathrm{free}}(\mathscr{G}_{k-1})|}{|\mathscr{A}\mathscr{P}_{\mathrm{forced}}(\mathscr{G}_{k-1})|} \right). \tag{15}$$

Theorem 5.12 (Epistemic-Physical Equivalence)**.** [Lean-verified bound.]

$$W_{\text{steer}}(\pi) \;\geq\; k_B T \cdot \Delta F[q], \tag{16}$$

where $\Delta F[q]$ is the reduction in variational free energy (in nats) achieved by π.

Proof. $\Delta F[q] = \sum_k \Delta q_k$ (nats). Cost per step $\geq k_B T \cdot \Delta q_k \cdot \ln 2$ by Theorem 5.10; $W_{\text{sel}} \geq 0$. Summing: $W_{\text{steer}} \geq k_B T \cdot \Delta F[q]$. □

The Action Value function of Section 6 is built directly on Theorems 5.10 and 5.12. The biological and evolutionary applications of these bounds are developed in RACI (Sandeman, 2026c).

6 The Mathematics of Graph Steering

6.1 Setup: Agency as Causal Forcing

This section derives the Action Value function for agentic graph steering directly from the Steering Cost Functional of RACI (Sandeman, 2026c). No additional assumptions are introduced beyond those already established in the RA suite.

Under passive RA-CSG dynamics, the adjacent possible $\mathscr{AP}_{\text{free}}(\mathscr{G}_n)$ at stage n contains all on-shell transitions available from the current graph configuration. An agent with a Tier 4 internal model intervenes by restricting the adjacent possible to a forced subset:

$$\mathscr{AP}_{\text{forced}}(\mathscr{G}_n) \;\subsetneq\; \mathscr{AP}_{\text{free}}(\mathscr{G}_n). \tag{17}$$

By Definition 5.11 (Steering Cost Functional, Section 5.4), the thermodynamic cost of a forced actualization sequence $\pi = (v_1, \ldots, v_k)$ is:

$$W_{\text{steer}}(\pi) \;:=\; \sum_{j=1}^{k} \Big[W_{\text{act}}(v_j) + W_{\text{sel}}(v_j) \Big], \tag{18}$$

where $W_{\text{act}}(v_j) = \Delta q_j \cdot k_B T \ln 2$ is the Landauer cost (Theorem 5.10) and $W_{\text{sel}}(v_j)$ is the selection cost (equation (15)).

By Theorem 5.12 (Epistemic-Physical Equivalence, Section 5.4):

$$W_{\text{steer}}(\pi) \;\geq\; k_B T \cdot \Delta F[q], \tag{19}$$

where $\Delta F[q]$ is the reduction in variational free energy (in nats) achieved by executing π. The factor $k_B T$ converts nats to joules.

6.2 The Three Components of Action Value

An agentic system selects the forced actualization sequence π^* that maximises a balance between expected epistemic gain, pragmatic benefit, and thermodynamic cost. We formalize this as the *Action Value* of a candidate sequence π given the current graph state $\mathscr{G}_n$:

Definition 6.1 (Action Value Function)**.** *For a candidate forced actualization sequence π executed from graph state $\mathscr{G}_n$, the Action Value is:*

$$V(\pi \mid \mathscr{G}_n) \;=\; \alpha \cdot I(\pi; \mathscr{G}_{n+1}) \;+\; \beta \cdot \mathbb{E}\big[U(\mathscr{G}_{n+1}) \mid \pi, \mathscr{G}_n\big] \;-\; \gamma \cdot W_{\text{steer}}(\pi), \tag{20}$$

where $\alpha, \beta, \gamma > 0$ are weighting parameters, and the three terms are defined below.

Term 1: Epistemic Forcing (I). $I(\pi; \mathscr{G}_{n+1})$ is the mutual information between executing π and the resulting graph state $\mathscr{G}_{n+1}$—the expected reduction in the agent's uncertainty about the future graph structure that results from executing π:

$$I(\pi; \mathscr{G}_{n+1}) \;=\; H(\mathscr{G}_{n+1}) - \mathbb{E}_{\pi}\big[H(\mathscr{G}_{n+1} \mid \pi)\big], \tag{21}$$

where H denotes Shannon entropy over the probability distribution on future graph states induced by the RA-CSG dynamics. This term is high when the agent's internal model $\mathscr{G}_{\text{model}}$ has high uncertainty about the future, and low when the future is already well-characterized.

RA derivation. The mutual information $I(\pi; \mathscr{G}_{n+1})$ is the information-theoretic dual of the relative entropy $S(\rho \| \sigma_0)$ that serves as the actualization criterion in RAQM and RAQI (Sandeman, 2026a,d). Executing π forces specific actualization events that increase $S(\rho \| \sigma_0)$ irreversibly, thereby reducing the agent's uncertainty about the graph state. The epistemic term is therefore the information gain from *deliberately triggering* the relative entropy increase that constitutes actualization—using the Kinematic Snap as an instrument of perception.

Term 2: Pragmatic Steering (U). $\mathbb{E}[U(\mathscr{G}_{n+1}) \mid \pi, \mathscr{G}_n]$ is the expected utility of the graph state resulting from π. The utility function U encodes the agent's autopoietic objective: maintaining the criticality parameter $\sigma \approx 1$. Formally:

$$U(\mathscr{G}_{n+1}) \;=\; -\big(\sigma(\mathscr{G}_{n+1}) - 1\big)^2 \;-\; \kappa_{\text{threat}} \cdot \mathbb{P}[\text{Markov blanket breach} \mid \mathscr{G}_{n+1}], \tag{22}$$

where the first term penalises deviation from the edge-of-chaos regime and the second term penalises configurations that threaten Markov closure. The agent selects π^* that steers $\mathscr{G}_{n+1}$ toward $\sigma \approx 1$ and away from Markov blanket instability.

RA derivation. The utility function (22) is not an exogenous preference imposed on the framework; it is the autopoietic objective derived from the definition of Tier 3 systems in Section 4. A system is autopoietic if and only if it maintains $\sigma \approx 1$. The utility function is therefore the formal expression of the Tier 3 survival condition, elevated to an explicit optimization target by the Tier 4 agent.

Term 3: Thermodynamic Cost (W_{steer}). $W_{\text{steer}}(\pi)$ is the steering cost functional (18), comprising the Landauer cost of actualization and the selection cost of restricting the adjacent possible. This term prevents the agent from pursuing unlimited graph forcing: by the Epistemic-Physical Equivalence bound (19), the thermodynamic cost of any forced sequence is bounded below by $k_B T \cdot \Delta F[q]$. An agent cannot force continuous, unlimited actualizations because its available free energy budget is finite.

Crucially, the cost term applies to *internal* simulation sequences as well as external actions. The generative motif (Section 4.4.1) computes predictions of $\mathscr{G}_{\text{env}}$ via real, on-shell actualization events within the agent's own graph. Each such internal simulation step is a forced actualization sequence π_{sim} with cost:

$$W_{\text{steer}}(\pi_{\text{sim}}) \;\geq\; k_B T \cdot \Delta F[q_{\text{sim}}], \tag{23}$$

where $\Delta F[q_{\text{sim}}]$ is the variational free energy reduced by running the simulation. The Action Value function therefore governs both physical actions (movements, communications, resource acquisition) and cognitive actions (simulating counterfactuals, updating the generative motif, planning). There is no free lunch in cognition: every thought that reduces uncertainty about the future pays the Landauer floor cost for the uncertainty it resolves. This directly accounts for the high metabolic cost of predictive neural processing.

Remark 6.2 (Correspondence with Friston's Expected Free Energy). The Action Value function (20) has a precise correspondence with Friston's Expected Free Energy $G(\pi)$ (Friston, 2010; Friston et al., 2017), which decomposes as:

$$G(\pi) \;=\; \underbrace{-I(\pi; \mathscr{G}_{n+1})}_{\text{epistemic value}} + \underbrace{\mathbb{E}\big[\log q(\mathscr{G}_{n+1} \mid \pi) - \log \tilde{p}(\mathscr{G}_{n+1})\big]}_{\text{risk (pragmatic cost)}}, \tag{24}$$

where $\tilde{p}$ is the agent's prior over preferred states. In Friston's framework, the agent selects the policy π that minimizes $G(\pi)$; in the ACD framework, the agent maximizes $V(\pi \mid \mathscr{G}_n)$.

The two are related by:

$$V(\pi \mid \mathscr{G}_n) \approx -G(\pi) - \gamma W_{\text{steer}}(\pi), \tag{25}$$

where the correspondence is:

- $\alpha I(\pi; \mathscr{G}_{n+1}) \leftrightarrow$ epistemic value in $-G(\pi)$: the expected reduction in uncertainty about the future graph state.
- $\beta\, \mathbb{E}[U(\mathscr{G}_{n+1})] \leftrightarrow$ pragmatic value $-\mathbb{E}[\log q - \log \tilde{p}]$: the expected alignment of the resulting state with the agent's preferred states (here, $\sigma \approx 1$ and Markov closure).
- $\gamma W_{\text{steer}}(\pi) \leftrightarrow$ the additional thermodynamic cost term, which is absent in the standard Free Energy Principle but is mandatory in the RA account: every forced actualization sequence—including internal simulation—pays the Landauer floor.

The ACD framework therefore provides the *physical substrate* for which the Free Energy Principle is the *functional description*: Friston's $G(\pi)$ is the information-theoretic shadow of the thermodynamic reality captured by $V(\pi \mid \mathscr{G}_n)$. The RA derivation adds one term that is invisible in the standard Free Energy Principle but physically non-negotiable: the Landauer cost γW_{steer}, which enforces the thermodynamic limits on cognitive capacity identified in Proposition 6.3.

6.3 The Optimal Steering Policy

The agent's optimal policy selects the forced actualization sequence π^* that maximises the Action Value:

$$\pi^* = \operatorname{argmax}_{\pi \in \Pi_{\text{admissible}}} V(\pi \mid \mathscr{G}_n), \tag{26}$$

where $\Pi_{\text{admissible}}$ is the set of all causal sequences available to the agent given its current Markov blanket boundary and free energy budget.

Several properties of (26) follow directly from the RA framework:

Proposition 6.3 (Thermodynamic Bound on Steering)**.** *For any admissible policy π,*

$$V(\pi \mid \mathscr{G}_n) \leq \alpha \cdot I(\pi; \mathscr{G}_{n+1}) + \beta \cdot \mathbb{E}[U(\mathscr{G}_{n+1})] - \gamma \cdot k_B T \cdot \Delta F[q(\pi)]. \tag{27}$$

The thermodynamic cost sets a lower bound on the free energy that must be expended to achieve any given epistemic or pragmatic gain.

Proof. Substitute the bound (19) into Definition 6.1. □

Corollary 6.4 (Finite Intelligence)**.** *No agent operating within a finite free energy budget can maximize both epistemic and pragmatic value simultaneously. The Action Value function encodes a thermodynamically enforced trade-off between knowing more (high I) and doing more (high U) that is fundamental to all intelligent systems, not a contingent architectural limitation.*

6.4 Surprise-Triggered Versus Value-Scored Forcing

Not all forced actualization sequences are selected by continuous optimization of (26). Some forcing events are triggered by threshold crossings in the agent's internal model. We distinguish two modes:

Definition 6.5 (Surprise-Triggered Forcing)**.** *A surprise-triggered forcing event occurs when the relative entropy of the agent's current state estimate against the expected state exceeds a threshold:*

$$S(\rho_{\text{current}} \| \rho_{\text{expected}}) > \theta_{\text{force}}, \tag{28}$$

at which point the agent executes the forced sequence π^*_{threat} that minimizes the probability of Markov blanket breach.

Definition 6.6 (Value-Scored Forcing)**.** *A value-scored forcing event is selected by continuous optimization of the Action Value function* (20) *across all admissible sequences, without a threshold trigger. This mode governs proactive, anticipatory steering when no immediate threat is detected.*

The two modes correspond to the fight-or-flight versus deliberative planning distinction in neuroscience, and to the distinction between reactive and proactive control in artificial systems. Both are instances of the same underlying optimization (26); the surprise trigger is a computationally efficient approximation that prioritizes immediate threat responses over full policy evaluation.

7 Empirical Predictions

The ACD framework makes several predictions that distinguish it from standard complex systems theory and are in principle testable.

P1: Causal depth of metabolic networks. By the linear scaling result of RACI (Sandeman, 2026c) ($A_{\mathrm{RA}}(M) \approx \kappa \cdot A(M)$ in condensed phases), evolutionary selection pressure should drive organisms toward biosynthetic pathways of minimal Assembly Index A. ACD predicts that across evolutionary lineages, organisms should exhibit systematic reduction in the Assembly Index of their critical molecules over evolutionary time, independently of thermodynamic cost. This is testable by comparing Assembly Index values across metabolic pathways in organisms at different evolutionary distances.

P2: Tier 3/4 transition and the internal model threshold. ACD predicts a sharp transition in information processing at the Tier 3/4 boundary: the emergence of a persistent internal model $\mathscr{G}_{\mathrm{model}}$ should correspond to a qualitative change in the agent's ability to perform counterfactual causal reasoning. Systems below the transition can react to environmental perturbations but cannot anticipate them. Systems above the transition can simulate counterfactual graph extensions and select actions that prevent perturbations before they arrive. This transition is testable in neuroscience via the presence or absence of prospective coding in neural activity.

P3: Thermodynamic cost of intelligence. Proposition 6.3 implies a strict thermodynamic cost for epistemic gain. ACD predicts that the metabolic cost of intelligent behavior scales with the mutual information acquired per unit time, with a proportionality constant set by $k_B T$. This is consistent with known metabolic scaling of neural tissue but provides a more precise bound: the free energy cost per bit of information acquired is at least $k_B T \ln 2$, the Landauer minimum.

P4: Optimal coarse-graining scale. The recursive hierarchy (Π_n) predicts that stable Markov blankets exist only at specific scales determined by the causal structure of the underlying graph. ACD predicts that biological and cognitive systems will exhibit characteristic scales at which information integration is maximized—corresponding to the scales at which the coarse-graining projection Π_n achieves maximum causal closure. This is testable via integrated information theory measures (Tononi, 2004) applied at multiple spatial and temporal scales.

8 Relationship to the RA Suite

The present paper extends the RA research program in a specific direction. Its relationship to the four companion papers is as follows:

RAQM (Sandeman, 2026a) Provides the foundational actualization criterion ($p^\mu p_\mu = m^2$, the Kinematic Snap) and the relative entropy criterion for actualization events. ACD uses these as the Level 0 building blocks of the recursive hierarchy.

RAGC (Sandeman, 2026b) Provides the Local Ledger Condition, the RA-CSG growth dynamics, and the adjacent possible formalism. ACD's coarse-graining projection Π_n is built on top of the CSG dynamics.

RACI (Sandeman, 2026c) Develops the biological and evolutionary applications of the theorems proved in Section 5 of the present paper. Specifically, RACI provides: the full C1′ variational argument and B3LYP/6-311+G* computational survey establishing the Assembly Index Correspondence; the detailed Causal Firewall derivation and edge-of-chaos phase transition; England compatibility and evolutionary predictions; the complete treatment of agency in biological context; and the four empirical predictions for astrobiology, evolutionary biology, and origin-of-life research. The division of labour is precise: RAHC proves the foundational theorems and establishes the universal hierarchy; RACI applies them to the specific domain of biological complexity and life.

RAQI (Sandeman, 2026d) Provides the Kinematic Coherence Bound and the connection between actualization timescales and information processing limits. ACD's prediction P3 (thermodynamic cost of intelligence) is a macroscopic extension of the RAQI bounds on quantum computing systems.

RADM (Sandeman, 2026e) Extends the actualization-density field to galactic scales. The critical actualization density λ_c at which the causal DAG undergoes dimensional reduction from $d_s = 3$ to $d_s = 2$ (Durhuus-Jonsson-Wheater theorem) is a cosmological instance of the same phase-transition mechanism that ACD identifies as a tier boundary in the complexity hierarchy. The sparse-graph regime ($\mu \ll 1$) in RADM is structurally analogous to the pre-Causal-Firewall abiotic chemistry regime in RACI — both involve actualization graphs below the threshold for stable self-referential structure.

EB (Sandeman, 2026g) Defines the Covariant Stochastic Graph Rewriting System whose Step 5 coarse-graining underlies Π_n. Recent progress: the BDG sign criterion ($S_{\mathrm{BDG}}(v) > 0$) as the discrete actualization filter yields the Poisson-CSG generative measure $c_k^{\mathrm{RA}}(x) \propto e^{-\mu}\mu^k$, making c_k local and environment-dependent — a structural advance that closes the measure problem for RA's CSG dynamics.

The eleven-paper suite also includes the particle classification and Standard Model topology paper (RATM (Sandeman, 2026RATM)), the field equation uniqueness paper (RACL (Sandeman, 2026RACL)), the Standard Model gauge structure and Koide formula paper (RASM (Sandeman, 2026h)), and the programme overview (Foundation (Sandeman, 2026i)).

9 Open Problems and Hard Walls

Several open problems remain before ACD becomes a complete theory.

Hard Wall 1: The Causal Firewall Limit Theorem — Resolved. The Tier 2/3 boundary corresponds to the critical condition $\sigma \equiv \lambda\, \tau_d\, \ell^3 = 1$ of RACI (Sandeman, 2026c). This is now proved rather than conjectured, via the identification of σ with the Poisson-CSG parameter $\mu = \lambda\, V_{\mathrm{coh}}\, p_{\mathrm{th}}$ (RAGC (Sandeman, 2026b)) and the application of the directed Erdős-Rényi theorem (Erdős & Rényi, 1960): a directed random causal graph with mean degree μ has a giant strongly connected component (shared causal ledger) if and only if $\mu > 1$. The Causal Firewall threshold $\sigma = 1$ is exactly this percolation transition. The trace-distance convergence bound between observer-dependent actualization histories scales as $\varepsilon \sim N^{-1/3}$ at criticality and decays exponentially above it. *Remaining gap:* deriving the identification $p_{\mathrm{th}} = \tau_d/\tau_{\mathrm{mol}}$ from the RA-CSG dynamics (a single calculation in open quantum systems theory). **Collaborator target:** open quantum systems, decoherence theory.

Hard Wall 2: Computational Tractability of Π_n. The coarse-graining projection Π_n is defined in terms of indistinguishable probability distributions over future macro-observables (equation (2)). Computing this projection for realistic systems is in general computationally intractable. Identifying efficiently computable approximations to Π_n that preserve the relevant causal structure is an open problem. **Collaborator target:** computational complexity theory, causal discovery algorithms.

Hard Wall 3: Quantitative Utility Function. The utility function $U(\mathscr{G}_{n+1})$ in (22) is defined qualitatively in terms of the criticality parameter σ and the Markov blanket breach probability. Making this quantitative requires explicit models of both quantities from the RA-CSG dynamics. The Causal Firewall conjecture provides a candidate expression for the threshold, but the full utility function requires the limit theorem of Hard Wall 1. **Collaborator target:** open quantum systems, mathematical biology.

Hard Wall 4: Connecting Π_n to Integrated Information. Prediction P4 connects ACD to integrated information theory (Tononi, 2004). Making this connection rigorous requires showing that the scales at which Π_n achieves maximum causal closure correspond to the scales at which integrated information Φ is maximized. This would provide ACD with a precise operational criterion for identifying the "natural" levels of description of a complex system. **Collaborator target:** information theory, consciousness research.

10 Conclusion

Algorithmic Causal Dynamics provides a unified physical account of the hierarchy of complexity. By grounding every tier—from on-shell boson exchanges to metabolic cells to reasoning minds—in the same formal mechanism of recursive causal closure, the conceptual boundaries between fundamental physics, biological autopoiesis, and cognition dissolve. Not because they are "really" the same thing, but because they are different regimes of the same topological process.

The key contributions of this paper are:

1. A rigorous topological definition of irreducible emergence as *causal closure*: a subgraph emerges at scale n when it achieves a stable Markov blanket and can be treated as a single effective causal node at scale $n+1$. This is the first such definition grounded in causal graph physics rather than information-theoretic postulates.
2. The recursive coarse-graining hierarchy $(\Pi_1, \Pi_2, \ldots, \Pi_n)$, which generates the full four-tier continuum of physical complexity—from on-shell actualization events through dissipative structures, autopoietic systems, and agentic intelligence—as successive regimes of causal closure produced by a single formal mechanism.
3. The Generative Motif definition and the Tier 4 Transition Criterion, providing the first precise topological criterion for distinguishing reactive autopoiesis from predictive agency in terms of internal causal graph structure.
4. A derivation of the Action Value function for agentic graph-steering directly from the RA thermodynamic framework, establishing the formal correspondence $V(\pi|\mathscr{G}_n) \approx -G(\pi) - \gamma W_{\text{steer}}(\pi)$ that positions RA as the physical substrate of Friston's Free Energy Principle.
5. Four empirical predictions that distinguish ACD from standard complex systems theory and connect to neuroscience, astrobiology, and integrated information theory.

The hierarchy is not a ladder climbing toward intelligence as its goal; it is a physical structure in which each tier is a stable regime of causal closure, as real and as fundamental as any other. Chemistry is not "merely" physics, nor is biology "merely" chemistry, nor cognition "merely" biology. Each is a distinct topological regime generated by the same underlying mechanism and irreducible to the tier below it—not because reductionism fails in principle, but because the macroscopic causal edges that govern behaviour at each tier are structurally absent at lower scales.

Intelligence, at the top of this hierarchy, is not mysterious and not special in kind. It is the tier at which causal closure becomes sufficiently deep and recurrent that a subgraph can model the causal structure of its environment, anticipate threats to its own Markov closure, and steer the growth of the universal DAG to ensure its continued actualization. The thermodynamic cost of this capacity—grounded in Proposition 6.3 and anchored empirically by the metabolic budget of the brain—is not an evolutionary inefficiency. It is the Landauer floor for running a physical simulation of a complex world. That the universe has generated systems willing to pay this cost is one of the most striking structural facts about the causal graph we inhabit.

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